



PURDUE POLYTECHNIC INSTITUTE
Department of Computer and Information Technology

CNIT 27200 – Phase 2: DDL and Data Load

Spring 2025

Due: Friday, April 11, 2025, by 11:59 pm via **Brightspace**

Instructions for Phase 2:

1. Read through the list of required documentation that will need to be created in this phase.
2. Use the checklist at the end of this document to verify that you have included all of the required items in your Phase 2 Project zip file.
 - a. **The TAs will be downloading your packets and running the scripts to test for accuracy.**

List of required documentation:

1. **DATA DEFINITION LANGUAGE** - Create a script for your database that includes creation of your database:
 - a. NOTE: See the following instructions for how to forward your ERD to a SQL script from SQL Developer.

Directions for forwarding the ERD to a .sql script file:

1. Open .dmd file and specifically open the Relational Model (the Physical model)
2. Pull down the File menu at the top, then Data Modeler -> Export -> DDL file
3. Leave the option at the top as Oracle and press the Generate button
4. Then press the OK button on the next screen.
5. Finally, save the document as a .sql file

You may need to make adjustments to your .sql file before loading it into Oracle. You must also add DROP TABLE CASCADE CONSTRAINTS commands to your script to drop each table before you create it. You each have your own Oracle accounts, so you should all load the same .sql file individually.

- b. Your initial forward engineer of the ERD to a DDL will create a single .sql script to implement all database objects, relationships, and permissions in your database.
 - i. See troubleshooting guide at the end of this document for help.

- c. Added Business Rule: Determine the fields in the database that are REQUIRED and update your data model accordingly (this doesn't change the design). At MINIMUM, make at least 3 non-keyed attributes required in your database. You can edit your generated script to make some non-keyed attributes not null.
2. **DATA MANIPULATION LANGUAGE** - Set up the test environment:
 - a. Set up a way to collaborate with your team the data you plan to use in each table. Suggestions: MS Teams, Google Docs, One Drive.
 - b. Create a script with the INSERT statements to load test data into your database
 - c. The test data sets up a controlled environment for testing.
 - d. Create at least 10 records for each of your entities.
 - e. If you find that some of the values are long, you may have to alter the datatype length of some of the attributes.
 - f. Some fields can be null, but keep the null values to a minimum. In at least one table, you will need to make some foreign keys null in a non-identifying relationship for the final phase (you will need to demonstrate a RIGHT JOIN).
 - g. The data needs to be varied and useful. You will be running queries against this data in the final phase. Use varied FK values from the parent PKs. For example, you should not just use one of the PK values for all 10 records in the child table FK. But do not use 10 different PK values in the FK. Use multiples of some, none of others.
 - h. **No Lorem ipsum. Use meaningful data.** Points will be deducted.
3. For the Phase 2 submission, combine the DDL and DML into a single .sql file that creates the database, loads data into the new database, reports the structure of each created table, and all data in each table using select *.
 - a. Submit a single **.sql file** sufficient to create and populate all tables.
 - i. Name this script document: **teamXX_milestone2.sql**
 - ii. In a comment section at the top, include the names of your team members.
 - iii. Do not forget to add the DROP TABLE xxxx CASCADE CONSTRAINTS commands at the beginning of the script for all of the tables in your database.
 - iv. The file should start with the database structure (DDL)
 - v. Then populate the database with insert into statements (DML)
 - vi. Add a Describe statement for each table in your database
 - vii. Run a Select * from *tablename* for each table in your database.
 - viii. Your execution should result in a clean install of the database into Oracle with all PK/FK relationships created.
 - ix. **Note:** be sure to include appropriate comments & formatting.
4. Submit a separate text document that reports the structure of each table in your new database (ie, I want a DESCRIBE for each table)
5. Submit a separate document with select * statements and results for each table in your database.
6. Read the Submission checklist below before submitting.

Phase 2 Submission Checklist:

1. There are 4 documents to submit in Phase 2. Zip the 4 documents into one file called **S25_M2.zip**
 - a. DOCUMENT 1: Submit an activity report

- i. Include a cover page with your CNIT 27200 Team Number within the course, DB name, team members, date and phase #.
 - ii. Table of Attendance chart – how and when did you meet? How long?
 - iii. Contribution Table
 - iv. Detailed Contribution Explanation (each student must explain their own contribution)
 - v. If your team made changes to the ERD, resubmit a readable copy.
- b. DOCUMENT 2: .sql script
 - i. Name this script document: **milestone2.sql**
 - ii. Should include DDL and DML commands, DESCRIBE statements, and Select * from *tablename* statements.
- c. DOCUMENT 3: DESCRIBE statements and results for each table in the DB (text file)
- d. DOCUMENT 4: SELECT * statements and results for each table in the DB (text file)

One person from your team should be designated to submit the zipped Phase 2 packet in Brightspace by April 11th at 11:59 pm. (Suggestion: Have that person notify all of you when the submission is complete)

We will test every team's script for accuracy.

Some advice for this phase:

1. After Lab 8 Part B, drop all of the tables from the Lunches DB and the Penguin DB.
 - a. This is important if any of your tables have the same name as the tables in these two Databases.
2. There is a specific order in which these tables must be created and populated in the script.
 - a. You must drop all of the tables with CASCADE CONSTRAINTS at the beginning of the script.
 - b. Then you should create each of the parent tables.
 - c. Then you can create the child tables based on the relationships. NOTE: You cannot set a foreign key reference to a table that has yet to be created.
 - i. Some students like to create all of the tables with the parent keys first, then set the foreign keys using ALTER TABLE for each child table. You still have to be careful with the order, but it can be somewhat easier to manage.
 - ii. Watch out for FK constraint names that are too long. They will error out.
 - d. Once all of the tables are created successfully, and foreign keys are also set up correctly, you can set the values for the parent tables.
 - e. After the parent tables are populated, you can follow the same order as above to populate the child tables.
3. Take special care to communicate with your team members on the values used for PKs and FKs. You cannot use values in the FKs that do not match the values in the corresponding PKs. **This error commonly happens when the team has poor communication.**
 - a. Some teams are using Microsoft Teams with success, some are using Google Docs, and some are using OneDrive. Find a cloud option that works for your team.

- b. The teams that have the least errors use a method for collaborating on what data will be used in their database.
- 4. Use varied data. Do not use the same value in every FK, nor should you use a different FK for every record.
 - a. For example, a project has a DEPARTMENT table and an EMPLOYEE table:
 - i. There should be 10 different departments and 10 different employees
 - ii. **DO NOT** – make all employees from the same department
 - iii. **DO NOT** – make all employees from a different department
 - iv. **DO** – make a few from one department, a few from another department, some from yet another department, and a couple with no department.
 - v. This way when you get to the final phase, you could run a COUNT on the number of employees per department. It would give you varied data (not 10 from one department, or 1 for 10 departments).
- 5. Use meaningful data. No Lorem ipsum. Think about the sample data from Phase 1 to guide you.
- 6. Stay in constant contact with your team, and make sure each person contributes to this phase. It is a lot of work.